

## CRESSIER, Switzerland

WMO: 066060

Lat: 47.0475N

Lon: 7.0592E

Elev: 1414

StdP: 13.96

Time Zone: 1.00 (EUC)

Period: 11-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 19.1       | 23.1 | 9.3                           | 9.3 | 19.1 | 17.3 | 13.8 | 25.6 | 25.7                  | 39.4 | 23.1 | 35.4 | 9.6                   | 50   | 0.488 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 20.0                   | 86.9            | 69.2 | 83.8 | 68.1 | 80.6 | 66.4 | 71.0                | 83.1 | 69.3 | 80.8 | 67.8 | 77.9 | 6.4                  | 210  |

| Dehumidification DP/MCDB and HR |       |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 67.4                            | 106.1 | 74.5 | 65.7 | 99.9 | 72.7 | 64.3 | 95.2 | 71.6 | 35.6          | 83.1 | 34.2 | 80.8 | 33.0 | 77.9 | 79.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |     |            |     |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|-----|------------|-----|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |     | n=50 years |     |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max | Min        | Max |      |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n) | (o)        | (p) |      |
| 22.0              | 18.8 | 16.0 | DB                         | 16.0 | 91.5               | 4.6 | 2.5                                                | 12.7 | 93.3       | 10.0 | 94.8       | 7.5 | 96.3       | 4.1 | 98.1 |
|                   |      |      | WB                         | 14.7 | 76.2               | 4.7 | 1.7                                                | 11.3 | 77.5       | 8.6  | 78.5       | 5.9 | 79.4       | 2.5 | 80.7 |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 51.3   | 35.9  | 35.6  | 43.6  | 50.9  | 56.5  | 64.8  | 68.0  | 67.3  | 60.3  | 51.9  | 42.6  | 37.3  |
|                                                                        | DBStd       | 13.02  | 6.21  | 6.88  | 5.56  | 6.18  | 6.09  | 5.73  | 5.23  | 5.44  | 4.97  | 5.43  | 5.35  | 5.58  |
|                                                                        | HDD50       | 1794   | 439   | 404   | 207   | 63    | 13    | 0     | 0     | 0     | 0     | 45    | 230   | 393   |
|                                                                        | HDD65       | 5317   | 903   | 824   | 664   | 424   | 271   | 72    | 30    | 39    | 156   | 405   | 672   | 857   |
|                                                                        | CDD50       | 2274   | 0     | 1     | 8     | 89    | 213   | 445   | 559   | 537   | 310   | 105   | 7     | 0     |
|                                                                        | CDD65       | 325    | 0     | 0     | 0     | 0     | 7     | 67    | 124   | 111   | 16    | 0     | 0     | 0     |
|                                                                        | CDH74       | 2892   | 0     | 0     | 0     | 10    | 82    | 621   | 1107  | 949   | 123   | 0     | 0     | 0     |
|                                                                        | CDH80       | 779    | 0     | 0     | 0     | 0     | 6     | 173   | 327   | 266   | 7     | 0     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 5.4    | 6.4   | 6.2   | 6.2   | 5.8   | 5.6   | 5.3   | 5.1   | 4.3   | 4.9   | 4.4   | 5.0   | 5.9   |
| (15) Precipitation                                                     | PrecAvg     | 47.7   | 3.5   | 3.4   | 3.1   | 3.6   | 4.4   | 4.3   | 4.5   | 4.6   | 3.9   | 4.1   | 4.1   | 4.3   |
|                                                                        | PrecMax     | 61.8   | 8.9   | 8.1   | 10.9  | 7.0   | 8.1   | 8.7   | 9.5   | 10.8  | 7.2   | 7.2   | 9.9   | 9.3   |
|                                                                        | PrecMin     | 34.3   | 0.8   | 1.0   | 0.9   | 0.8   | 2.0   | 2.3   | 2.0   | 0.9   | 1.5   | 0.7   | 0.4   | 0.1   |
|                                                                        | PrecStd     | 7.0    | 2.1   | 1.8   | 2.3   | 1.9   | 1.8   | 1.7   | 1.7   | 2.0   | 1.7   | 1.7   | 2.3   | 2.2   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 52.5  | 58.0  | 65.8  | 75.5  | 80.1  | 89.0  | 91.1  | 89.4  | 81.3  | 70.1  | 60.8  | 52.8  |
|                                                                        |             | MCWB   | 47.8  | 48.5  | 51.8  | 59.5  | 64.0  | 71.7  | 70.1  | 69.9  | 65.4  | 58.8  | 53.5  | 49.4  |
|                                                                        | 2%          | DB     | 49.5  | 52.0  | 62.4  | 70.9  | 76.2  | 85.0  | 87.2  | 86.2  | 78.0  | 67.1  | 57.2  | 50.4  |
|                                                                        |             | MCWB   | 46.4  | 45.8  | 50.6  | 56.5  | 61.5  | 69.6  | 68.4  | 69.4  | 65.3  | 58.6  | 52.9  | 47.9  |
|                                                                        | 5%          | DB     | 46.6  | 48.3  | 58.6  | 67.7  | 72.7  | 80.9  | 83.6  | 83.0  | 74.2  | 63.9  | 54.0  | 48.7  |
|                                                                        |             | MCWB   | 43.6  | 43.6  | 48.6  | 55.0  | 59.7  | 67.1  | 67.4  | 68.3  | 63.7  | 57.4  | 50.8  | 46.4  |
|                                                                        | 10%         | DB     | 44.1  | 45.0  | 54.6  | 63.8  | 69.1  | 77.0  | 80.3  | 79.5  | 71.0  | 61.1  | 51.1  | 45.9  |
|                                                                        |             | MCWB   | 41.9  | 41.2  | 46.8  | 53.1  | 58.2  | 64.8  | 65.8  | 66.4  | 62.0  | 56.4  | 48.7  | 43.8  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 49.5  | 49.4  | 52.9  | 59.9  | 65.7  | 74.5  | 72.2  | 72.4  | 69.1  | 62.2  | 55.8  | 50.2  |
|                                                                        |             | MCDB   | 51.2  | 54.6  | 63.4  | 72.0  | 77.0  | 87.0  | 85.4  | 83.4  | 74.8  | 65.7  | 58.6  | 51.1  |
|                                                                        | 2%          | WB     | 46.8  | 46.3  | 51.0  | 57.1  | 62.8  | 70.8  | 70.2  | 70.6  | 66.7  | 60.1  | 53.5  | 48.5  |
|                                                                        |             | MCDB   | 49.1  | 51.6  | 60.6  | 68.2  | 72.5  | 82.7  | 83.4  | 82.5  | 73.7  | 63.9  | 56.4  | 50.1  |
|                                                                        | 5%          | WB     | 44.1  | 44.2  | 49.2  | 55.1  | 60.4  | 68.6  | 68.5  | 69.1  | 65.2  | 58.7  | 51.2  | 46.7  |
|                                                                        |             | MCDB   | 46.4  | 48.4  | 57.4  | 64.7  | 69.3  | 79.1  | 80.2  | 80.2  | 72.1  | 62.2  | 53.7  | 48.5  |
|                                                                        | 10%         | WB     | 42.1  | 41.6  | 47.3  | 53.2  | 58.5  | 66.3  | 67.1  | 67.7  | 63.4  | 57.1  | 48.9  | 44.2  |
|                                                                        |             | MCDB   | 44.0  | 44.8  | 53.7  | 61.2  | 66.2  | 75.1  | 77.4  | 77.4  | 69.6  | 60.5  | 50.5  | 45.6  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 8.7   | 11.6  | 16.6  | 18.4  | 17.6  | 18.7  | 20.0  | 20.3  | 18.0  | 14.5  | 9.5   | 8.3   |
|                                                                        |             | MCDBR  | 12.7  | 20.0  | 26.0  | 27.0  | 25.0  | 25.8  | 26.5  | 25.6  | 23.4  | 19.9  | 14.9  | 11.8  |
|                                                                        |             | MCWBR  | 9.6   | 13.5  | 14.7  | 13.5  | 11.3  | 11.8  | 10.7  | 10.5  | 12.4  | 12.8  | 10.9  | 9.8   |
|                                                                        | 5% WB       | MCDBR  | 12.2  | 18.4  | 23.2  | 23.8  | 21.3  | 24.3  | 23.6  | 23.7  | 21.3  | 16.8  | 13.8  | 11.0  |
|                                                                        |             | MCWBR  | 9.7   | 13.2  | 13.8  | 12.5  | 10.2  | 11.8  | 10.3  | 10.4  | 12.3  | 11.8  | 10.7  | 9.6   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.299 | 0.325 | 0.362 | 0.401 | 0.410 | 0.413 | 0.412 | 0.403 | 0.381 | 0.359 | 0.323 | 0.297 |
|                                                                        | taud        |        | 2.433 | 2.363 | 2.314 | 2.232 | 2.252 | 2.290 | 2.287 | 2.323 | 2.364 | 2.410 | 2.453 | 2.441 |
|                                                                        | Ebn at Noon |        | 252   | 266   | 272   | 270   | 271   | 271   | 269   | 267   | 262   | 250   | 239   | 238   |
|                                                                        | Edh at Noon |        | 22    | 28    | 35    | 41    | 42    | 41    | 40    | 37    | 33    | 27    | 21    | 19    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 384   | 652   | 1027  | 1381  | 1592  | 1885  | 1826  | 1588  | 1223  | 778   | 424   | 312   |
|                                                                        | RadStd      |        | 48    | 99    | 154   | 202   | 149   | 175   | 160   | 147   | 112   | 76    | 54    | 55    |

## Historical Trends

|                          | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|--------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                          |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| Station Only             | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Regional (108 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
|                          | N/A   | N/A     | N/A    | +1.31   | +0.72 | +0.54 | N/A         | N/A   | +95   | +31   |

Nomenclature: See separate page